	Reference No.: BatStateU-FO-COL-03	Effectivity Date: January 3, 2017	Revision No.: 00

Title:	**TOPIC ABSTRACT FORM**
College	College of Informatics and Computing Sciences
Proponent/s	
Program	Bachelor of Science in Computer Science

Proposed Title: Real-Time Mosquito Habitat Detection on Android Devices Using Optimized Lightweight CNN Models in Tropical Urban Environments.

Background of the Proposed Topic:

Mosquito-borne diseases such as dengue, malaria, Zika virus, and West Nile virus are a constant threat to the public. According to the World Health Organization (WHO), since the 1960s, it was known from fewer than 10 countries where it is endemic and only a few thousand cases were reported per year; now it has ballooned to 128 endemic countries and is estimated to have 400 million infections and 100 million symptomatic cases per year. Local programs such as the ReMoVE dengue have detected that the virus is in circulation in Quezon City in the barangay of Old Balara (Heralde et. al, 2023).

Traditional monitoring, such as ovitraps and larval surveys, relies heavily on constant supervision and depends on barangay health workers and volunteers. Obra et. al (2022) detected Aedes aegypti (Yellow fever mosquito) and Aedes Albopictus (Asian tiger mosquito) through the routine monitoring of ovitrap in Quezon City. In Egypt, Abo El Ela et. al (2023) found and recorded nine different species of mosquitoes across polluted and clean bodies of water, with Culex pipiens (Common house mosquito) being the most prevalent. In Malaysia, Fikri et al. (2025) analyzed 6,027 water containers and discovered that 6.7% of them are positive for larvae; they also identified that plastic, rubber, and cement containers are the highest risks. This shows that doing manual surveys is helpful, but it is hard to maintain and is in dire need of automated methods.

The recent advances in AI have made it possible to have compact and solid Convolutional Neural Networks (CNNs). CNNs enable efficient image recognition capabilities using low memory and processing costs. Fariza, Yuwono, Akbar, Asmara, and Aryawan (2023) developed a mobile larvae recognition system, and they compared the two: TensorFlow Serving and TensorFlow Lite. They successfully reached 96.1 percent accuracy and showed that on-device classification is possible even for household use. The use of CNNs can also be extended for mosquito habitat detection. Passos et al. (2020) trained their models using aerial videos of possible mosquito habitats; this reported an F1 score of 0.65 for tires and 0.77 for water tanks. Prachyabrued et al. (2020) applied CNNs using Google Street View images and identified containers in urban neighborhoods. Elango et al. (2022) trained YOLO variants, YOLOv4, YOLOR, and YOLOv5, using satellite imagery to detect large bodies of water. The 3 YOLO variants averaged an IoU of 55.85, with YOLOv4 getting the highest accuracy. Ananthajothy et al. (2023) compared CNNs with few-shot learning using drone imagery; the CNNs outperformed FSL with 84.29% accuracy after only training with 100 epochs. FSL, on the other hand, only showed 79% after training the model with 2000 total epochs.

Further research has integrated CNN-based detection that has its public health applications. Yin et al. (2021) showed that adding container details from street-view images improved the dengue-risk predictions compared to the models that have no container details. Knoblauch et al. (2024) have extended this approach by linking the CNN-based image detection from satellite and street-view imagery to ovitrap egg and larva data. This created detailed Aedes aegypti hotspot maps. This shows that image-based detection enhances habitat detection and risk mapping.

Recent deep learning methods enhanced the localization accuracy and the robustness of the model. Using a new aerial image dataset, Mylvaganam et al. (2025) found that Masked R-CNNs outperformed the selective search CNNs. Achieving 85% accuracy in identifying possible mosquito breeding sites like tires and cans, even under challenging and noisy conditions. This demonstrates its suitability to be deployed in the field for mosquito habitat monitoring.

CNNs have also proven reliable for species and larvae detection. Nolte et al. (2025) classified mosquito species using wing images, achieving 98.3 percent accuracy across 21 taxa. Amiruddin and Kadir (2020) compared CNN architectures, with MobileNetV2 reaching 80 percent accuracy, reinforcing its suitability for lightweight deployment. Kulasekara (2024) applied multispectral drone imagery to larvae detection, reaching 95 percent accuracy, while stage classification achieved 73 percent. These studies confirm CNN adaptability across tasks, devices, and conditions.

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The availability of datasets is continuously growing and makes it possible to advance model training and its development. The MosquitoFusion dataset contains 1,204 labeled images of mosquitoes, swarms, and mosquito habitats. Mosquito Breeding Habitats vol. 3 dataset, on the other hand, offers 1194 images that feature stagnant water, puddles, and vegetation. The Vision and Text-Based Mosquito Breeding Detection dataset is helpful for both object detection, like YOLO, and vision-language models like CLIP. The Mosquito Breeding Sites Computer Vision adds another 112 images of common breeding sites like tires, buckets, puddles, etc.

These studies collectively reveal that there is a gap in the current literature. Existing dengue surveillance relies heavily on manual labor and manpower despite the evidence that shows mosquitoes thrive in a wide range of environments. CNNs demonstrated that it is reliable and showed a strong performance in detecting mosquito species, habitats, and larvae. It also shows that it is possible for mosquito habitat detection to come to mobile devices. However, most existing studies depend heavily on open datasets that usually come from non-tropical environments and/or large-scale imagery, which limits the applicability of the models to the conditions in the Philippines.

To address this limitation, this study will augment open-source datasets with locally captured images of common possible mosquito breeding containers from Filipino households and urban environments rather than solely relying on training with the open-source datasets. The hybrid approach is adopted to achieve a good balance of variety of datasets, realism, and efficiency. The research benchmarks **MobileNetV3**, **EfficientNet-Lite**, and **Tiny-YOLO** to find which model offers the best balance between accuracy, speed, and size for real-world use. This study presents a practical yet low-cost framework for real-time mosquito habitat detection. This study supports SDG 3: Good Health and Well-Being through early mosquito-borne disease detection and strengthens local surveillance.

Objectives of the Study

General Objective

To benchmark and optimize a CNN model for real-time mosquito habitat detection using mobile devices, reducing required manpower, labor, and time for monitoring, and enabling rapid public health response.

Specific Objectives

1. To collect and preprocess at least 1,000 labeled images of possible breeding sites of mosquitoes (tires, buckets, puddles, stagnant water, etc.) for model training.
2. To train and evaluate MobileNetV3, EfficientNet-Lite, and Tiny-YOLO from the acquired dataset to determine which model achieves a minimum 85% accuracy in classifying potential breeding sites.
3. To compare models in terms of their accuracy, latency (inference target is 300 ms per image for mid-range android devices, model size (target is <25 MB), and storage requirements to identify the most efficient architecture for mobile among the 3.
4. To apply pruning and quantization to the best-performing model and measure the changes in its accuracy, latency, and size. Targeting at least a 30% reduction in size but losing no more than 5% in accuracy.
5. To deploy the optimized model into a fully working Android application that is capable of real-time detection of possible mosquito habitats, with detection feedback display within 1 second.

Significance of the Study

Dengue continues to affect millions across the Philippines. Many households still depend on manual checks to find mosquito breeding sites, which are often slow and inconsistent. This study builds a lightweight CNN-based system that detects potential mosquito habitats directly on Android devices. The goal is to make real-time detection faster, low-cost, and accessible without internet or specialized tools.

The system benefits several groups:

- **Households and Communities.** You receive instant detection results for containers that may breed mosquitoes. This supports the DOH “search and destroy” campaign and lowers dengue risk at home.
  Public Health Workers. You reduce the time spent on field inspections. The tool assists in quick habitat verification and allows you to focus on control and prevention work.
  Local Governments and Health Agencies. You gain data to identify which containers are most common in local households. This improves surveillance and planning for dengue prevention.
  Future Researchers. You access the dataset, model card, and prototype for continued work in AI-driven public health applications.

The study introduces three key novelties. First, it supplements open-source datasets with **locally captured mosquito habitat images** from Filipino homes and urban areas to improve contextual accuracy. Second, it **benchmarks multiple lightweight CNNs** MobileNetV3, EfficientNet-Lite, and Tiny-YOLO under realistic mobile conditions. Third, it **analyzes the trade-off between accuracy, latency, and model size** to determine the most efficient model for real-world use.

Unlike prior studies that relied solely on large-scale aerial or non-tropical datasets, this research focuses on **on-device mosquito habitat detection tailored for Philippine environments**. It introduces a mobile-optimized and locally informed approach to support early dengue intervention. This study contributes new practical knowledge by offering a **deployable, data-driven, and low-cost framework** for community-level mosquito surveillance aligned with SDG 3: Good Health and Well-Being.

Methodology:

This study follows this process to properly address, benchmark, and optimize the 3 CNN models selected for mosquito habitat detection using mobile phones. The steps here cover the following: dataset preparation, model training, optimization, and mobile deployment.

1. **Data Acquisition and Labeling**
   The dataset that will be used in this study will be a mix of open-source datasets from Kaggle, Roboflow, and will be supplemented by locally captured images. This study will use the Mosquito Breeding Habitats Vol. 3 dataset with 1,194 images, the MosquitoFusion dataset with 1204 images, and the Mosquito Breeding Sites Computer Vision dataset with 112 images. These 3 open-sourced datasets contain the following images: stagnant water, puddles, containers, and vegetation. The images are captured using different lighting and weather conditions, and each image includes bounding boxes that are in YOLO or Pascal VOC format, labeling objects like buckets, tires, water containers, etc.

   To increase the relevance and locality of the images to the Philippines, about 500 to 700 locally captured images will be supplemented and added to the dataset. The images will be taken from the households of the Batangas area using Android devices with at least 12 - 48 MP cameras. The images will be from indoor and outdoor scenes across different weather conditions, lighting, angles, and time of day. Photos will highlight and focus on containers and their surroundings without identifiable features. The images will be labeled using LabelImg or Roboflow Annotate; it will then be tagged for the presence of water, if the object is covered or uncovered, and the lighting conditions.

   The final dataset will have about 2,800 to 3,000 labeled images that will be split into 70-15-15. 70% of the dataset will be for model training, 15% is for validation, and the other 15% is for testing. All images will be resized to a uniform resolution and will be normalized between the open and locally sourced images.

   Data augmentation techniques will be used, including adjustment of the brightness, rotations, and cropping. This is to reduce overfitting and make the dataset more varied. The image resolutions will be 224x224 for MobileNet and EfficientNet and 416x416 for Tiny-YOLO. This is to enable faster processing and minimize the needed memory on mobile devices.

- **Data Preprocessing**
  The images in the dataset will be resized, normalized, and stored. The augmented images will be merged with the original images. This way, every type of possible habitat will be fairly represented, even if they have different lighting conditions.
- **Model Selection and Justification**
  Three (3) lightweight CNN models are selected for their efficiency in mobile inference:
  • MobileNetV3 for low latency and compact design
  • EfficientNet-Lite for balanced accuracy and performance
  • Tiny-YOLO for high-speed object detection
- **Model Training and Evaluation**
  The models will be trained using TensorFlow and PyTorch frameworks. Transfer learning from ImageNet-pretrained weights will be used to shorten training time. Each of the models will run for 50 to 100 epochs/rounds using the Adam Optimizer with learning rate scheduling to adjust learning speed as training progresses.

  The 3 models will use the identical dataset splits, augmentations, and batch sizes for a fair comparison. The model's performance will be measured using accuracy, precision, recall, F1-score, mean average precision (mAP) at 0.5 IoU, inference latency, model size, and memory usage.

This study will benchmark its results against prior literature. Fariza et al (2023) achieved 96.1% accuracy in larvae classification, and Passos et al. (2020) achieved F1 scores of 0.65 and 0.77 for their aerial habitat detection. These studies will serve as a baseline reference.

5. **Model Optimization**
   The best performing model among the 3 will be optimized using pruning and quantization. Pruning will remove the unnecessary and redundant connections, reducing its size. Quantization will convert 32-bit weights to 8-bit weights, thus reducing memory use and making computation faster.

   Performance will be measured using:

   CR = (Original Size − Optimized Size) / Original Size × 100%

   AR = (Optimized Accuracy / Baseline Accuracy) × 100%

   The goal in this stage is a 30 percent reduction in model size with minimal loss in accuracy(>5%).

6. **Conversion and Mobile Deployment**
   The optimized model will then be converted to TensorFlow Lite (.tflite) format so that it can run using Android devices. It will then be built into an Android Studio application using Java. The application will let the user either take a picture or choose from the gallery to perform real-time detection. On the screen, it will draw bounding boxes and labels if it is a potential habitat and breeding site for mosquitoes.
7. **On-Device Testing and Performance Measurement**
   Testing will be done on a commercially available mid-range Android device. The test will include 100 locally captured images with different lighting and background conditions. Metrics will include inference latency, model size, and runtime memory use.

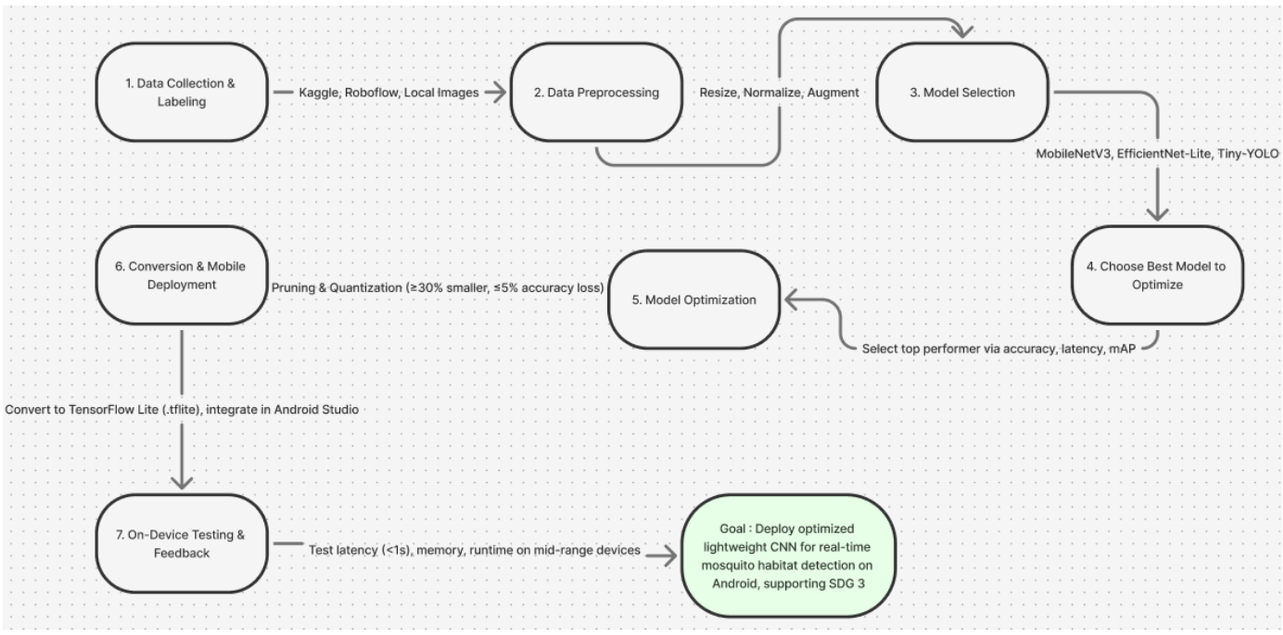
   Latency will be measured as:

   Latency = Total Inference Time / Number of Images

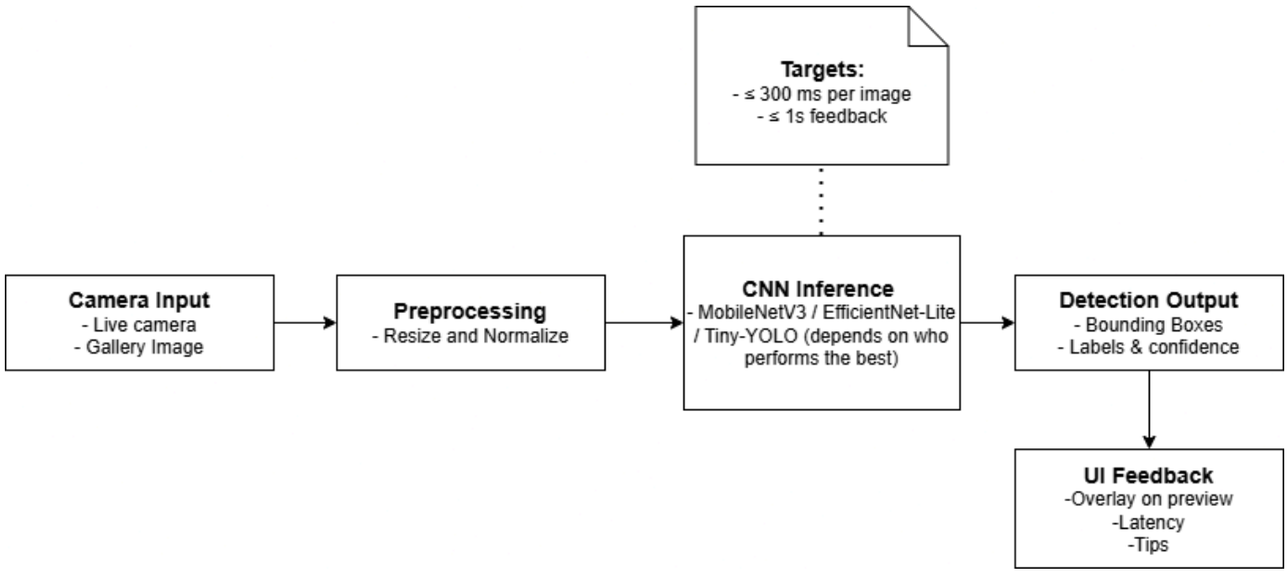
   The goal is 300 ms (milliseconds) per image with detection and feedback within 1 second.

   Confusion matrices are going to be used to detect classification errors, while ROC and Precision-Recall plots will show the balance between its sensitivity and specificity. Performance summary will highlight the trade-offs of the model's size, latency, and accuracy.

   This testing will focus on the reliability of the mobile inference under field conditions like crowded or chaotic environments, partially blocked objects, and poor lighting to ensure that the model will be able to handle real-world use cases.



Concept Diagram



Deployment

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TOPIC EVALUATION COMMITTEE

The topic abstract has been thoroughly reviewed by the Topic Evaluation Committee.

	Signature	**Remarks**
College Dean	Dr. Princess Marie B. Melo	
Department Chair	Ms. Lanie P. Palad	
Faculty Expert/s	Mr. Raymond Kit Rodriguez	

1. Title Evaluation

   - The title still lacks geographic or environmental specificity (e.g., *"in the Philippines"* or *"in tropical environments"*).

   - Add local or environmental context to reflect the study setting.
   Example: "Real-Time Mosquito Habitat Detection on Android Devices Using Optimized Lightweight CNN Models in Tropical Urban Environments."

2. Novelty and Contribution

   - The originality of the study is well-stated, but it would benefit from a concise statement highlighting the new dataset.

   - Add a one-sentence summary in the *Significance* or *Conclusion* section emphasizing that the locally captured dataset is the study's primary novel contribution.

3. Methodology Evaluation

   - Include a conceptual workflow diagram that visually presents the step-by-step process from data collection to mobile deployment.

4. Dataset Discussion

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- Add a summary table showing each dataset used, number of images, type of data (open-source or local), and purpose (training, validation, or testing).

5. Benchmarking and Model Comparison

   - Provide a comparative summary table that outlines the baseline studies, their reported results, and the study's target metrics for clearer benchmarking.

6. Evaluation and Testing

   - Specify the exact test device model and specifications used for performance testing to enhance transparency and reproducibility.

7. Simulation and Deployment

   - The description of the deployment process is well-written but would benefit from visual support.

   - Add a system flow diagram showing the entire process — from camera input → CNN inference → detection output → user interface feedback.

Comments and Recommendations

1. Title Evaluation

- The title *"Optimizing Lightweight CNN Algorithms for Real-Time Mosquito Habitat Detection on Mobile Devices"* is clear and direct.
- It accurately indicates the use of lightweight CNNs and mobile implementation.
- However, it sounds generic and lacks geographic or contextual focus.
- The title suggests an optimization contribution but doesn't specify the method or innovation type.

Recommendations:

- Emphasize what is unique about your study — local dataset, deployment context, or optimization approach.
- Possible improved versions:
  - "Optimization of Lightweight CNNs for On-Device Mosquito Habitat Detection in Tropical Environments."
  - "Real-Time Mosquito Habitat Detection on Android Devices Using Optimized Lightweight CNN Models."
- Ensure that the title reflects both **content and novelty**.

2. Problem–Objective Alignment

- The problem focuses on the need for rapid mosquito habitat detection to support public health monitoring.
- The objectives highlight model optimization, benchmarking, and deployment.
- The direct connection between each problem and its corresponding objective is weakly expressed.
- The objectives are measurable but lack target metrics such as latency thresholds or target accuracy.

Recommendations:

- Explicitly match every identified problem with a specific objective.
- Clarify what each objective will achieve in measurable terms (for example, reducing model size by a specific percentage or achieving real-time detection within a defined time frame).
- Include one overall success indicator that defines when the objective is considered met.

3. Novelty and Contribution

- The proposed models and optimization techniques (MobileNet, EfficientNet-Lite, Tiny-YOLO, pruning, quantization) are widely known and used.
- The novelty is not yet strongly demonstrated.
- The abstract briefly mentions using local data, which could be the study's most original component, but this is underdeveloped.

Recommendations:

- Clearly identify the **main novelty**, such as:
  - The creation of a locally captured mosquito habitat image dataset.

    - The evaluation of multiple CNNs under realistic mobile conditions.
    - The integration of accuracy–latency–size trade-off analysis for mobile devices.
- State explicitly how your approach differs from existing studies in the same field.
- Include a short summary in your paper that emphasizes the new knowledge or practical benefit the work will contribute.

4. Methodology Evaluation

- The methodology lacks detailed steps and is too high-level.
- It mentions training and optimization but omits precise processes, algorithms, or frameworks.
- No visual workflow or process explanation is given.

Recommendations:

- Present the methodology as a clear **sequence of steps**:
    - Data acquisition and labeling.
    - Model selection and justification.
    - Model training using transfer learning or fine-tuning.
    - Model optimization through pruning and quantization.
    - Conversion to mobile-compatible format (such as TensorFlow Lite).
    - Testing on mobile hardware and performance measurement.
- Describe each step's purpose and how it directly addresses one or more objectives.
- Include a **conceptual diagram** illustrating data flow, model processing, and output delivery.
- Mention any software tools or frameworks (ex.:, TensorFlow, PyTorch, Android Studio).
- Include short mathematical expressions to define performance measures or optimization ratios.

5. Dataset Discussion

- The datasets are listed but not described in depth.
- There is no mention of image quantity, annotation type, or image conditions.
- The combination of multiple datasets with local data raises questions of consistency and quality.

Recommendations:

- Describe the dataset composition in narrative form:
    - Mention where the data come from, what scenes or objects are captured, and how they were labeled.
    - Explain the variety of environments (indoor/outdoor, lighting, container types, water visibility).
    - Describe how local data collection was conducted, including devices, locations, and consent or permissions.
    - Explain how data will be divided for training, validation, and testing.
- Mention whether data augmentation will be used and why.
- Clarify how the dataset supports real-time detection on mobile devices (ex.:, small, efficient images).

6. Benchmarking and Model Comparison

- Benchmarking is mentioned but lacks structure and context.
- It is unclear how models will be compared or what metrics will be prioritized.
- There are no references to baseline studies for comparison.

Recommendations:

- Define the performance indicators clearly — such as accuracy, F1-score, precision, recall, mean average precision (mAP), inference speed, model size, and memory usage.
- State how each model will be tested under similar conditions to ensure fair comparison.
- Explain what baseline models or existing works you will benchmark against and why they are chosen.
- Discuss the expected trade-off between accuracy and computational efficiency.

7. Evaluation and Testing

- The evaluation criteria are briefly listed but not explained.

- The testing environment (hardware, conditions, datasets) is not described.
- The plan to measure latency or performance on mobile devices is missing.

Recommendations:

- Provide a clear explanation of how each metric will be computed.
- Describe the test setup, including device type, hardware specs, and software environment.
- Discuss how the testing will reflect real-world conditions (lighting, occlusion, motion).
- Include plans for result interpretation (ex.:, confusion analysis, ROC curves, performance summary).
- Emphasize real-time performance validation on mobile devices rather than only on a desktop GPU.

8. Simulation and Deployment

- Simulation details are vague, with no mention of environment, tools, or performance measurement.
- The mobile deployment plan is not broken down into steps.

Recommendations:

- Describe the simulation environment (ex.:, Android emulator, real mobile devices, TensorFlow Lite Benchmark Tool).
- Specify how input (camera or image file) will be tested in real time.
- Discuss measurement of processing speed, energy use, and responsiveness.
- Outline how the system will handle real-world variations (ex.:, lighting, motion, background noise).
- Add a brief conceptual diagram showing the entire flow: from camera input → CNN inference → detection output → user interface.

9. Overall Observations

- The topic is highly relevant and aligns with global and local public health needs.
- The current proposal is promising but lacks depth in experimental design and justification of novelty.
- There is limited evidence of benchmarking or systematic performance analysis.

Recommendations:

- Strengthen the logical flow: *Problem → Objectives → Methodology → Evaluation → Results → Impact.*
- Include a closing paragraph summarizing contributions, potential impact, and limitations.
- Explicitly connect every major section back to the main objective of creating a lightweight, mobile-ready CNN solution for mosquito habitat detection.
- Make the methodology and evaluation sections more detailed and visual with process diagrams and metric equations.